

# **Networked Influence in a Tabula Rasa Legislature**

Co-authorship, Ideological Dynamics, and Political Success  
in the Chilean Constitutional Convention (2021–2022)

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## A natural quasi-experiment

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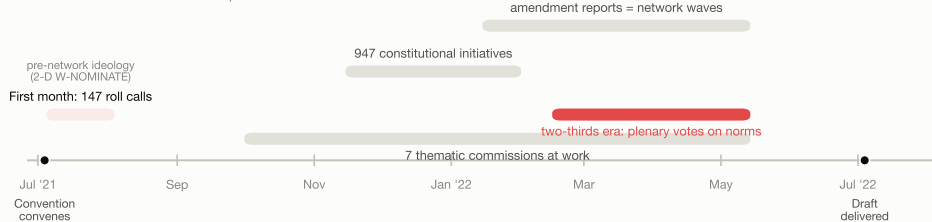
- In an ordinary legislature, today's collaboration network is the sediment of decades: careers, committees, party discipline. Any estimate of “what organizes collaboration” is contaminated by that history.
- The Chilean Constitutional Convention (Jul 2021 – Jul 2022) started from **zero relational stock**: a majority of newcomers and independents, 17 reserved indigenous seats, a body that dissolved on delivery — no electoral shadow of the future.
- Two rules structure everything. Each constitutional initiative required **8–16 sponsors**: signing = forming a visible, dated coalition. Each norm needed **two thirds of the floor (103 of 154)**: being born was cheap, surviving was expensive.

A note on standpoint: I am a methodologist coming from the natural sciences. I report what the models find and comment on some of it; the political reading is ongoing joint work with Jorge Fábrega (sociologist, UDD).

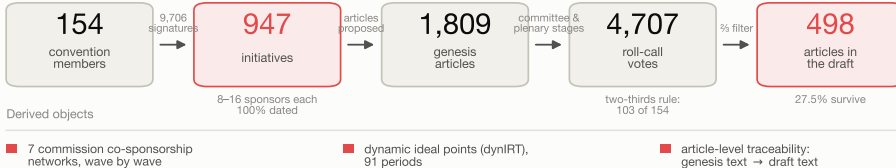
# The data: one year, fully linked

## One year, fully dated

Chilean Constitutional Convention, Jul 2021 – Jul 2022



## The pipeline: every signature, article and vote, linked



## The seven thematic commissions

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|    | Name (short)       | Members | % lawyers | % exper. | Age  | Degree | Initiatives | Waves | Amendments | Multi-si |
|----|--------------------|---------|-----------|----------|------|--------|-------------|-------|------------|----------|
| C1 | Political system   | 25      | 60        | 36       | 42.2 | 1.44   | 89          | 4     | 365        | 266      |
| C2 | Const. principles  | 18      | 22        | 22       | 45.8 | 1.28   | 114         | 3     | 53         | 1        |
| C3 | Form of the state  | 30      | 37        | 40       | 45.1 | 1.43   | 51          | 6     | 241        | 212      |
| C4 | Fundamental rights | 30      | 30        | 13       | 45.9 | 1.03   | 283         | 5     | 395        | 154      |
| C5 | Environment        | 19      | 21        | 16       | 44.7 | 1.26   | 133         | 5     | 559        | 0        |
| C6 | Justice systems    | 17      | 88        | 18       | 42.6 | 1.65   | 93          | 6     | 381        | 317      |
| C7 | Knowledge systems  | 15      | 13        | 0        | 51.7 | 1.33   | 64          | 8     | 332        | 54       |

- Composition varies wildly across commissions — C6 is 88% lawyers, C7 has no politically experienced member. **Every model below is estimated within commission.**

## Pre-network ideology: measured before the network existed

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- **Pre-network positions:** 2-D W-NOMINATE on the *first month* of floor votes only (147 roll calls, Jul–Aug 2021), replicating Fábrega (2022).
  - That window predates the commissions (Oct), the initiatives (Nov) and the operative two-thirds rule (Feb) — so  $\theta$  is exogenous to the network we study.
  - Why 2-D: the second dimension separates the reserved indigenous seats from the classic left–right cleavage; correct classification rises from 89.4% to 91.6%.
- **Dynamic revealed positions:** dynamic IRT over all 4,707 roll calls  $\rightarrow$  trajectories  $\theta_{i,t}$  over 91 periods,

$$\Pr(\text{yes}_{iv}) = \Phi(\beta_v \theta_{i,t(v)} - \alpha_v), \quad \theta_{i,t} \sim \mathcal{N}(\theta_{i,t-1}, \omega^2).$$

## Research questions

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- **RQ1 — Formation**: with zero relational stock, could the co-sponsorship network be predicted from what people *brought with them* (ideology, bloc, district, profile)?
- **RQ2a — Positions**: does network exposure *move* members' ideological positions, or do members select into like-minded coalitions?
- **RQ2b — Behavior**: does voting *defection* travel along co-sponsorship ties?
- **RQ3 — Success**: what makes an article survive into the draft — and does the *context* an initiative is born into matter?

RQ1 — Formation

Could we predict the network?

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## The tool: a hybrid bipartite ERGM, by commission and by bloc

- Unit of analysis: the **person**  $\times$  **initiative** signature ( $y_{ia} = 1$  if member  $i$  signed initiative  $a$ ). No projection — one signature counts once.

$$\Pr(Y = y) = \frac{\exp(\theta^\top s(y))}{\kappa(\theta)}, \quad s(y) = \underbrace{\left( \sum_{i,a} y_{ia} \right)}_{\text{edges}}, \underbrace{\left( \sum_a \sum_{i < j} y_{ia} y_{ja} \mathbf{1}\{b_i, b_j\} = \{g, h\} \right)}_{\text{bloc mixing: 15 cells}}, \underbrace{\left( \sum_a \sum_{i < j} y_{ia} y_{ja} \mathbf{1}[x_{ia} = x_{ja}] \right)}_{\text{profile homophily}}$$

- Separating by bloc **never cuts the network** (that would mutilate coalitions): the network stays whole and the *counters* specialize by bloc. Each homophily counter partitions exactly into 5 within-bloc + 1 across-bloc terms.
- Plus: per-document ranges of  $\theta_1, \theta_2$ , age, education; the bloc contingent's own  $\theta_1$  range; gwb1degree; gwdsp. Seven models, one per commission (pooling inverts homophily signs — Simpson's paradox, verified).
- Estimation: MPLE as a logistic on change statistics (certified against `ergm` to  $10^{-11}$ ); SEs by **initiative bootstrap** ( $B = 500$ : resample initiatives, rebuild network, re-estimate).



| Same status, by bloc of the pair        | C1                 | C2                 | C3         | C4                 | C5                 | C6                 | C7                 |
|-----------------------------------------|--------------------|--------------------|------------|--------------------|--------------------|--------------------|--------------------|
| <b>— Lawyer</b>                         |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | +0.11              | +0.11              | +0.13      | −0.03              | +0.15*             | −0.26              | −0.00              |
| Right                                   | −0.11              | −0.15**            | −0.01      | +0.01              | +0.10              | +0.13*             | −0.06              |
| Left                                    | −0.07              | +0.06              | +0.02      | +0.06 <sup>+</sup> | +0.00              | +0.11*             | −0.07              |
| Local lists                             | +0.12              | −0.11              | −0.09      | +0.09 <sup>+</sup> | −0.00              | −0.13              | −0.15              |
| Reserved seats                          | +0.06              | +0.24              | +0.10      | −0.26 <sup>+</sup> | +0.20*             | −0.21              | −0.04              |
| Across blocs                            | +0.06              | −0.08*             | +0.02      | +0.04*             | −0.05 <sup>+</sup> | +0.07 <sup>+</sup> | +0.06 <sup>+</sup> |
| <b>— Prior institutional experience</b> |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | +0.11              | −0.03              | +0.09      | +0.09**            | −0.09              | −0.01              | +0.12              |
| Right                                   | −0.13              | −0.14              | −0.03      | +0.05              | −0.14              | +0.00              | −0.11              |
| Left                                    | +0.12 <sup>+</sup> | +0.13*             | +0.09      | +0.11**            | +0.04 <sup>+</sup> | +0.06              | +0.00              |
| Local lists                             | −0.25              | −0.26              | [+11.1***] | +0.05              | [+8.1***]          | −0.20              | −0.30              |
| Reserved seats                          | +0.06              | +0.00              | −0.08      | −0.05              | −0.11              | −0.10              | −0.05              |
| Across blocs                            | +0.04              | +0.06 <sup>+</sup> | +0.03      | +0.06**            | +0.04*             | +0.04              | +0.04              |
| <b>— Gender</b>                         |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | −0.08              | −0.10              | −0.07      | +0.00              | +0.02              | −0.02              | +0.05              |
| Right                                   | +0.14*             | +0.03              | +0.05      | −0.06              | +0.04              | −0.03              | +0.09              |
| Left                                    | −0.12              | −0.03              | −0.06      | −0.03              | −0.02              | −0.04              | −0.09              |
| Local lists                             | −0.21              | −0.26              | −0.61      | −0.20**            | −0.38*             | +0.19              | −0.37              |
| Reserved seats                          | −0.11              | −0.50              | −1.77      | −0.60**            | −1.64***           | +0.04              | −0.26              |
| Across blocs                            | +0.04              | +0.12**            | +0.20*     | +0.13***           | +0.12***           | +0.08 <sup>+</sup> | +0.11*             |
| <b>— District</b>                       |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | −0.09              | −0.80              | −0.48      | +0.73*             | +0.18              | −0.74              | +0.30              |
| Right                                   | +1.00***           | +1.36***           | +0.42      | +0.41***           | +0.08              | +0.23              | +0.88*             |
| Left                                    | −0.64*             | −0.48*             | +0.45      | −0.45***           | −0.29              | −0.45*             | −0.04              |
| Local lists                             | +1.98 <sup>+</sup> | EMPTY              | −1.92**    | EMPTY              | −0.29              | +0.26              | EMPTY              |
| Reserved seats                          | −0.07              | +0.13              | −0.03      | −0.27 <sup>+</sup> | +0.30              | −0.03              | +0.19              |
| Across blocs                            | +0.54***           | +0.28*             | +0.67**    | +0.39***           | +0.47***           | +0.34*             | +0.46**            |

| Same status, by bloc of the pair        | C1                 | C2                 | C3         | C4                 | C5                 | C6                 | C7                 |
|-----------------------------------------|--------------------|--------------------|------------|--------------------|--------------------|--------------------|--------------------|
| <b>— Lawyer</b>                         |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | +0.11              | +0.11              | +0.13      | -0.03              | +0.15*             | -0.26              | -0.00              |
| Right                                   | -0.11              | -0.15**            | -0.01      | +0.01              | +0.10              | +0.13*             | -0.06              |
| Left                                    | -0.07              | +0.06              | +0.02      | +0.06 <sup>+</sup> | +0.00              | +0.11*             | -0.07              |
| Local lists                             | +0.12              | -0.11              | -0.09      | +0.09 <sup>+</sup> | -0.00              | -0.13              | -0.15              |
| Reserved seats                          | +0.06              | +0.24              | +0.10      | -0.26 <sup>+</sup> | +0.20*             | -0.21              | -0.04              |
| Across blocs                            | +0.06              | -0.08*             | +0.02      | +0.04*             | -0.05 <sup>+</sup> | +0.07 <sup>+</sup> | +0.06 <sup>+</sup> |
| <b>— Prior institutional experience</b> |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | +0.11              | -0.03              | +0.09      | +0.09**            | -0.09              | -0.01              | +0.12              |
| Right                                   | -0.13              | -0.14              | -0.03      | +0.05              | -0.14              | +0.00              | -0.11              |
| Left                                    | +0.12 <sup>+</sup> | +0.13*             | +0.09      | +0.11**            | +0.04 <sup>+</sup> | +0.06              | +0.00              |
| Local lists                             | -0.25              | -0.26              | [+11.1***] | +0.05              | [+8.1***]          | -0.20              | -0.30              |
| Reserved seats                          | +0.06              | +0.00              | -0.08      | -0.05              | -0.11              | -0.10              | -0.05              |
| Across blocs                            | +0.04              | +0.06 <sup>+</sup> | +0.03      | +0.06**            | +0.04*             | +0.04              | +0.04              |
| <b>— Gender</b>                         |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | -0.08              | -0.10              | -0.07      | +0.00              | +0.02              | -0.02              | +0.05              |
| Right                                   | +0.14*             | +0.03              | +0.05      | -0.06              | +0.04              | -0.03              | +0.09              |
| Left                                    | -0.12              | -0.03              | -0.06      | -0.03              | -0.02              | -0.04              | -0.09              |
| Local lists                             | -0.21              | -0.26              | -0.61      | -0.26**            | -0.36*             | +0.19              | -0.37              |
| Reserved seats                          | -0.11              | -0.50              | -1.77      | -0.60**            | -1.64***           | +0.04              | -0.26              |
| Across blocs                            | +0.04              | +0.12**            | +0.20*     | +0.13***           | +0.12***           | +0.08 <sup>+</sup> | +0.11*             |
| <b>— District</b>                       |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | -0.09              | -0.80              | -0.48      | +0.73*             | +0.18              | -0.74              | +0.30              |
| Right                                   | +1.00***           | +1.36***           | +0.42      | +0.41***           | +0.08              | +0.23              | +0.88*             |
| Left                                    | -0.64*             | -0.48*             | +0.45      | -0.45***           | -0.29              | -0.45*             | -0.04              |
| Local lists                             | +1.98 <sup>+</sup> | EMPTY              | -1.92**    | EMPTY              | -0.29              | +0.26              | EMPTY              |
| Reserved seats                          | -0.07              | +0.13              | -0.03      | -0.27 <sup>+</sup> | +0.30              | -0.03              | +0.19              |
| Across blocs                            | +0.54***           | +0.28*             | +0.67**    | +0.39***           | +0.47***           | +0.34*             | +0.46**            |

| Same status, by bloc of the pair        | C1                 | C2                 | C3         | C4                 | C5                 | C6                 | C7                 |
|-----------------------------------------|--------------------|--------------------|------------|--------------------|--------------------|--------------------|--------------------|
| <b>Lawyer</b>                           |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | +0.11              | +0.11              | +0.13      | -0.03              | +0.15*             | -0.26              | -0.00              |
| Right                                   | -0.11              | -0.15**            | -0.01      | +0.01              | +0.10              | +0.13*             | -0.06              |
| Left                                    | -0.07              | +0.06              | +0.02      | +0.06 <sup>+</sup> | +0.00              | +0.11*             | -0.07              |
| Local lists                             | +0.12              | -0.11              | -0.09      | +0.09 <sup>+</sup> | -0.00              | -0.13              | -0.15              |
| Reserved seats                          | +0.06              | +0.24              | +0.10      | -0.26 <sup>+</sup> | +0.20*             | -0.21              | -0.04              |
| Across blocs                            | +0.06              | -0.08*             | +0.02      | +0.04*             | -0.05 <sup>+</sup> | +0.07 <sup>+</sup> | +0.06 <sup>+</sup> |
| <b>— Prior institutional experience</b> |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | +0.11              | -0.03              | +0.09      | +0.09**            | -0.09              | -0.01              | +0.12              |
| Right                                   | -0.13              | -0.14              | -0.03      | +0.05              | -0.14              | +0.00              | -0.11              |
| Left                                    | +0.12 <sup>+</sup> | +0.13*             | +0.09      | +0.11**            | +0.04 <sup>+</sup> | +0.06              | +0.00              |
| Local lists                             | -0.25              | -0.26              | [+11.1***] | +0.05              | [+8.1***]          | -0.20              | -0.30              |
| Reserved seats                          | +0.06              | +0.00              | -0.08      | -0.05              | -0.11              | -0.10              | -0.05              |
| Across blocs                            | +0.04              | +0.06 <sup>+</sup> | +0.03      | +0.06**            | +0.04*             | +0.04              | +0.04              |
| <b>— Gender</b>                         |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | -0.08              | -0.10              | -0.07      | +0.00              | +0.02              | -0.02              | +0.05              |
| Right                                   | +0.14*             | +0.03              | +0.05      | -0.06              | +0.04              | -0.03              | +0.09              |
| Left                                    | -0.12              | -0.03              | -0.06      | -0.03              | -0.02              | -0.04              | -0.09              |
| Local lists                             | -0.21              | -0.26              | -0.61      | -0.20**            | -0.38*             | +0.19              | -0.37              |
| Reserved seats                          | -0.11              | -0.50              | -1.77      | -0.60**            | -1.64***           | +0.04              | -0.26              |
| Across blocs                            | +0.04              | +0.12**            | +0.20*     | +0.13***           | +0.12***           | +0.08 <sup>+</sup> | +0.11*             |
| <b>— District</b>                       |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | -0.09              | -0.80              | -0.48      | +0.73*             | +0.18              | -0.74              | +0.30              |
| Right                                   | +1.00***           | +1.36***           | +0.42      | +0.41***           | +0.08              | +0.23              | +0.88*             |
| Left                                    | -0.64*             | -0.48*             | +0.45      | -0.45***           | -0.29              | -0.45*             | -0.04              |
| Local lists                             | +1.98 <sup>+</sup> | EMPTY              | -1.92**    | EMPTY              | -0.29              | +0.26              | EMPTY              |
| Reserved seats                          | -0.07              | +0.13              | -0.03      | -0.27 <sup>+</sup> | +0.30              | -0.03              | +0.19              |
| Across blocs                            | +0.54***           | +0.28*             | +0.67**    | +0.39***           | +0.47***           | +0.34*             | +0.46**            |

| Same status, by bloc of the pair        | C1                 | C2                 | C3         | C4                 | C5                 | C6                 | C7                 |
|-----------------------------------------|--------------------|--------------------|------------|--------------------|--------------------|--------------------|--------------------|
| <b>— Lawyer</b>                         |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | +0.11              | +0.11              | +0.13      | -0.03              | +0.15*             | -0.26              | -0.00              |
| Right                                   | -0.11              | -0.15**            | -0.01      | +0.01              | +0.10              | +0.13*             | -0.06              |
| Left                                    | -0.07              | +0.06              | +0.02      | +0.06 <sup>+</sup> | +0.00              | +0.11*             | -0.07              |
| Local lists                             | +0.12              | -0.11              | -0.09      | +0.09 <sup>+</sup> | -0.00              | -0.13              | -0.15              |
| Reserved seats                          | +0.06              | +0.24              | +0.10      | -0.26 <sup>+</sup> | +0.20*             | -0.21              | -0.04              |
| Across blocs                            | +0.06              | -0.08*             | +0.02      | +0.04*             | -0.05 <sup>+</sup> | +0.07 <sup>+</sup> | +0.06 <sup>+</sup> |
| <b>— Prior institutional experience</b> |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | +0.11              | -0.03              | +0.09      | +0.09**            | -0.09              | -0.01              | +0.12              |
| Right                                   | -0.13              | -0.14              | -0.03      | +0.05              | -0.14              | +0.00              | -0.11              |
| Left                                    | +0.12 <sup>+</sup> | +0.13*             | +0.09      | +0.11**            | +0.04 <sup>+</sup> | +0.06              | +0.00              |
| Local lists                             | -0.25              | -0.26              | [+11.1***] | +0.05              | [+8.1***]          | -0.20              | -0.30              |
| Reserved seats                          | +0.06              | +0.00              | -0.08      | -0.05              | -0.11              | -0.10              | -0.05              |
| Across blocs                            | +0.04              | +0.06 <sup>+</sup> | +0.03      | +0.06**            | +0.04*             | +0.04              | +0.04              |
| <b>— Gender</b>                         |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | -0.08              | -0.10              | -0.07      | +0.00              | +0.02              | -0.02              | +0.05              |
| Right                                   | +0.14*             | +0.03              | +0.05      | -0.06              | +0.04              | -0.03              | +0.09              |
| Left                                    | -0.12              | -0.03              | -0.06      | -0.03              | -0.02              | -0.04              | -0.09              |
| Local lists                             | -0.21              | -0.26              | -0.61      | -0.20**            | -0.38*             | +0.19              | -0.37              |
| Reserved seats                          | -0.11              | -0.50              | -1.77      | -0.60**            | -1.64***           | +0.04              | -0.26              |
| Across blocs                            | +0.04              | +0.12**            | +0.20*     | +0.13***           | +0.12***           | +0.08 <sup>+</sup> | +0.11*             |
| <b>— District</b>                       |                    |                    |            |                    |                    |                    |                    |
| Center-left                             | -0.09              | -0.80              | -0.48      | +0.73*             | +0.18              | -0.74              | +0.30              |
| Right                                   | +1.00***           | +1.36***           | +0.42      | +0.41***           | +0.08              | +0.23              | +0.88*             |
| Left                                    | -0.64*             | -0.48*             | +0.45      | -0.45***           | -0.29              | -0.45*             | -0.04              |
| Local lists                             | +1.98 <sup>+</sup> | EMPTY              | -1.92**    | EMPTY              | -0.29              | +0.26              | EMPTY              |
| Reserved seats                          | -0.07              | +0.13              | -0.03      | -0.27 <sup>+</sup> | +0.30              | -0.03              | +0.19              |
| Across blocs                            | +0.54***           | +0.28*             | +0.67**    | +0.39***           | +0.47***           | +0.34*             | +0.46**            |

|                                                                       | C1       | C2                 | C3       | C4       | C5                 | C6                 | C7       |
|-----------------------------------------------------------------------|----------|--------------------|----------|----------|--------------------|--------------------|----------|
| <b>— Ideological range of the bloc's own contingent, per document</b> |          |                    |          |          |                    |                    |          |
| Center-left                                                           | -1.76    | +0.23              | +0.23    | +0.16    | -2.07              | +3.35 <sup>+</sup> | -1.54    |
| Right                                                                 | +0.20    | -0.30              | -5.29    | -5.33*   | -6.92              | -0.45              | -3.03    |
| Left                                                                  | -1.21    | -1.16              | -2.75    | -3.70*** | -3.81***           | +0.40              | -1.80    |
| Local lists                                                           | -4.82    | -2.71 <sup>+</sup> | +0.55    | -2.03**  | -2.81***           | -4.73*             | -0.92    |
| Reserved seats                                                        | -0.87    | +0.94              | +3.26    | -1.40    | -5.43 <sup>+</sup> | +0.15              | +0.93    |
| <b>— Controls and structure</b>                                       |          |                    |          |          |                    |                    |          |
| Edges (baseline)                                                      | +0.16    | -0.49              | -0.19    | -0.28    | -0.68              | -0.55              | -0.77    |
| Commission member                                                     | +0.79*** | +1.60***           | +0.87*** | +1.06*** | +1.66***           | +1.96***           | +2.20*** |
| Range of $\theta_1$ per document (global)                             | -2.06*** | -2.51***           | -3.85**  | -3.07*** | -0.09              | -1.89*             | -1.55    |
| Range of $\theta_2$ per document                                      | -2.10*** | -1.49*             | -1.73    | -2.35*** | -2.02**            | -1.49 <sup>+</sup> | -0.71    |
| Range of age per document                                             | -0.26    | +0.16              | -0.20    | +0.03    | -0.57**            | +0.01              | +0.17    |
| Range of education per document                                       | -0.27    | -0.57**            | -0.28    | +0.04    | -0.39*             | -0.33              | -0.02    |
| Serial signers (gwb1degree)                                           | -0.63    | -0.85              | -0.05    | -6.94*   | -1.88***           | +1.22*             | -0.55    |
| Repeated pairs (gwdsp)                                                | -0.42*** | -0.28***           | -0.38*** | -0.16*** | -0.23***           | -0.35***           | -0.24*** |

|                                                                       | C1       | C2                 | C3       | C4       | C5                 | C6                 | C7       |
|-----------------------------------------------------------------------|----------|--------------------|----------|----------|--------------------|--------------------|----------|
| <b>— Ideological range of the bloc's own contingent, per document</b> |          |                    |          |          |                    |                    |          |
| Center-left                                                           | -1.76    | +0.23              | +0.23    | +0.16    | -2.07              | +3.35 <sup>+</sup> | -1.54    |
| Right                                                                 | +0.20    | -0.30              | -5.29    | -5.33*   | -6.92              | -0.45              | -3.03    |
| Left                                                                  | -1.21    | -1.16              | -2.75    | -3.70*** | -3.81***           | +0.40              | -1.80    |
| Local lists                                                           | -4.82    | -2.71 <sup>+</sup> | +0.55    | -2.03**  | -2.81***           | -4.73*             | -0.92    |
| Reserved seats                                                        | -0.87    | +0.94              | +3.26    | -1.40    | -5.43 <sup>+</sup> | +0.15              | +0.93    |
| <b>— Controls and structure</b>                                       |          |                    |          |          |                    |                    |          |
| Edges (baseline)                                                      | +0.16    | -0.49              | -0.19    | -0.28    | -0.68              | -0.55              | -0.77    |
| Commission member                                                     | +0.79*** | +1.60***           | +0.87*** | +1.06*** | +1.66***           | +1.96***           | +2.20*** |
| Range of $\theta_1$ per document (global)                             | -2.06*** | -2.51***           | -3.85**  | -3.07*** | -0.09              | -1.89*             | -1.55    |
| Range of $\theta_2$ per document                                      | -2.10*** | -1.49*             | -1.73    | -2.35*** | -2.02**            | -1.49 <sup>+</sup> | -0.71    |
| Range of age per document                                             | -0.26    | +0.16              | -0.20    | +0.03    | -0.57**            | +0.01              | +0.17    |
| Range of education per document                                       | -0.27    | -0.57**            | -0.28    | +0.04    | -0.39*             | -0.33              | -0.02    |
| Serial signers (gwb1degree)                                           | -0.63    | -0.85              | -0.05    | -6.94*   | -1.88***           | +1.22*             | -0.55    |
| Repeated pairs (gwdsp)                                                | -0.42*** | -0.28***           | -0.38*** | -0.16*** | -0.23***           | -0.35***           | -0.24*** |

RQ2 — People

Does the network move positions (RQ2a) or behavior (RQ2b)?

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## RQ2a design: exposure and within-person change

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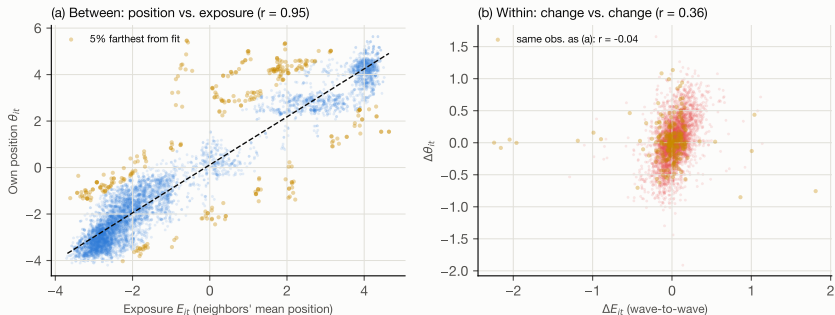
- Exposure of  $i$  at wave  $t$ : mean position of co-signers, weighted by accumulated collaboration  $w_{ij,t}$ ; the model uses only *within-person* variation:

$$E_{i,t} = \frac{\sum_{j \neq i} w_{ij,t} \theta_{j,t}}{\sum_{j \neq i} w_{ij,t}} \quad \Delta\theta_{i,t} = \alpha_i + \beta \theta_{i,t-1} + \lambda E_{i,t-1} + \varepsilon_{it}$$

- Individual fixed effects  $\alpha_i$ ; SEs clustered by member; 4,355 person-wave observations.
- If the network drags positions,  $\lambda > 0$ . Result:  $\hat{\lambda} = +0.007$  (SE 0.004) — **null**.



## Selection, not influence — in one picture



(a) Between persons: own position and neighborhood position correlate 0.95 — we choose neighborhoods that already resemble us. (b) Within person: if influence existed, the cloud would slope upward — the farther my neighborhood, the more I should move toward it. It does not ( $r$  flat); the amber 5% (the only people with room to be dragged) show  $r = -0.04$ .

## Exposure with temporal decay: a hint, and its shadow

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|                                               | $\lambda_w = 0.25$ | $\lambda_w = 0.50$ | $\lambda_w = 0.75$ |
|-----------------------------------------------|--------------------|--------------------|--------------------|
| Lagged exposure $\hat{\lambda}$               | +0.008             | +0.008             | +0.007             |
| SE (clustered); $p$                           | 0.004; .051        | 0.004; .042        | 0.004; .078        |
| Mean reversion $\hat{\beta} (\theta_{i,t-1})$ | -0.663***          | -0.663***          | -0.663***          |
| SE (clustered)                                | 0.024              | 0.024              | 0.024              |
| N (person-waves)                              | 4,355              | 4,355              | 4,355              |
| $R^2$ within                                  | 0.353              | 0.353              | 0.353              |

- With memory that discounts the past, the lag grazes significance.
- But the falsification fires too: *future* exposure “predicts” today’s change (+0.010,  $p = .030$ ) — the signature of selection, not influence.

## A dated exception: the norms era, measured with its own thermometer

- Clean test: decompose future exposure into its predictable part and its *innovation*  $\tilde{E}_{i,t+1}$  (the within-residual of lead on lag), and race them:

$$\Delta\theta_{i,t} = \alpha_i + \beta \theta_{i,t-1} + \lambda_{\text{lag}} E_{i,t-1} + \lambda_{\text{innov}} \tilde{E}_{i,t+1} + \varepsilon_{it}$$

| Thermometer                                       | Lag $\lambda_{\text{lag}}$ | Innovation $\lambda_{\text{innov}}$ | N     |
|---------------------------------------------------|----------------------------|-------------------------------------|-------|
| Standard $\theta$ (all 4,707 votes)               | +0.002 ( $p = .71$ )       | +0.062**                            | 3,328 |
| Regime-homogeneous $\theta$ (two-thirds era only) | + <b>0.016</b> **          | +0.053*                             | 3,038 |

- Survives measurement propagation (50 replicate re-fits of the era IRT; Rubin total SE  $\Rightarrow z = 2.53$ ) and concentrates at the *onset* of the two-thirds regime (Feb–Mar +0.021; Apr–May +0.007).
- Reading: selection is the rule; a bounded, dated accommodation appears when the 103-vote arithmetic was new. Lag and innovation *coexist* — not clean causality.

## RQ2b: defection travels along co-sponsorship ties

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- Defection: in roll call  $v$ , voting against the modal vote of one's own bloc.  
Exposure: the  $w$ -weighted share of  $i$ 's co-signers defecting *in the same roll call*:

$$X_{iv} = \frac{\sum_{j \neq i} w_{ij} D_{jv}}{\sum_{j \neq i} w_{ij}}, \quad \Pr(D_{iv} = 1) = \Lambda(\eta_i + \mu_v + \phi X_{iv})$$

---

|                                                                          | $\phi$          | $p$          |
|--------------------------------------------------------------------------|-----------------|--------------|
| Exposure to defectors (observed)                                         | +11.2           | $< 10^{-10}$ |
| Expected mechanically (permutation within bloc $\times$ vote, 200 draws) | +6.0 [p95: 6.1] | $< 0.005$    |

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- Half the raw effect is mechanical; **what remains is large**, travels more strongly *across* commissions, and leaves a short echo that dies within days. Carried by the *newcomer* pairs, not the old guard.

RQ3 — Texts

What makes an article survive?

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## The survival model: article-level, complete

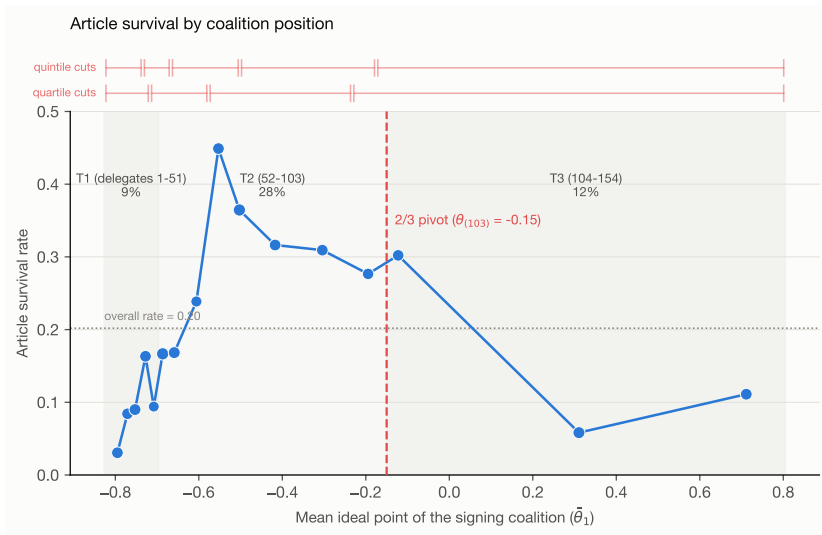
- Unit: the *article* (1,565 with sponsor coalition  $S_a$  of 2–16); 20.2% reach the draft.

$$\Pr(\text{survives}_a) = \Lambda\left(\alpha_c + \gamma_1 |\bar{\theta}_{1,S_a} - \theta_{1,(103)}| + \gamma_2 \text{sd}(\theta_{1,S_a}) + \gamma_3 |S_a| + \gamma_4^\top C_{S_a} + \gamma_5^\top H_{S_a}\right)$$

|                                                 | (1) Pivotal          | (2) + Network        | (3) + Human capital     |
|-------------------------------------------------|----------------------|----------------------|-------------------------|
| Coalition's distance to the $\frac{2}{3}$ pivot | -0.35 ( $p = .001$ ) | -0.48 ( $p = .009$ ) | -0.61 ( $p = .002$ )    |
| Ideological heterogeneity $\text{sd}(\theta_1)$ | +0.27 ( $p = .008$ ) | +0.46 ( $p = .001$ ) | +0.53 ( $p < 10^{-4}$ ) |
| Coalition size                                  | +0.20 ( $p = .12$ )  | +0.18 ( $p = .12$ )  | +0.15 ( $p = .21$ )     |
| Mean betweenness                                |                      | -0.22 ( $p = .083$ ) | -0.23 ( $p = .076$ )    |
| Mean constraint                                 |                      | +0.01 ( $p = .97$ )  | +0.26 ( $p = .28$ )     |
| Internal density (pairs with history)           |                      | +0.25 ( $p = .024$ ) | +0.32 ( $p = .002$ )    |
| Share of lawyers                                |                      |                      | -0.09 ( $p = .58$ )     |
| Share with prior experience                     |                      |                      | -0.16 ( $p = .44$ )     |
| Mean academic degree                            |                      |                      | -0.06 ( $p = .67$ )     |
| AIC                                             | 1393                 | 1384                 | 1386                    |

Articles win by geometry (near the pivot, ideologically wide) and by teams with internal history — not by credentials.

# Survival by coalition position: the arithmetic of 103



The two-thirds rule fixed a pivot at  $\theta_{1,(103)} = -0.15$ . The premium for widening lives in left coalitions

## Does the context an initiative is born into matter?

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- Member-level success:  $y_i$  = mean textual retention of the articles  $i$  co-signed. It clusters on the co-sponsorship network (Moran's  $I = 0.44$ ). The spatial Durbin model asks whether that context matters:

$$y = \rho W y + X\beta + WX\gamma + \varepsilon, \quad W = \text{row-normalized co-sponsorship network}$$

- $\rho = 0.891$  ( $p < 10^{-15}$ ; robust 0.63–0.95 across  $W$  variants). So “context matters” — **which by itself explains nothing**. The question is **what the context is**.
- Decomposition: the neighborhood's success  $Wy$  is predictable from one's *own* covariates ( $R^2 = 0.81$ ); and the only covariate that matters — distance to the  $\frac{2}{3}$  pivot — **migrates entirely from the person to the neighborhood** when moving from OLS to the Durbin model (own:  $p = .95$ ; neighborhood's:  $-0.27, p < .05$ ).
- Answer: **the context is the coalition you stand in** — its composition and its position in the coalition landscape. Success is a coalition good, not diffusion;  $\rho$  is read as clustering, not influence



## Takeaways

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- The network **was predictable** from pre-existing traits: bloc, 2-D ideology, and — conditionally — territory. Credentials organized nothing, in any bloc.
- The bloc partition pays: territory is the right's internal glue and the system's bridge; gender crosses blocs; experience lives in the left;  $gwdsp < 0$  everywhere — the structural signature of a tabula rasa.
- Positions barely move: **selection**, with one dated exception at the onset of the two-thirds regime.
- Behavior does travel: defection clusters along co-sponsorship ties at twice the mechanical rate, carried by newcomers.
- Texts win by **geometry and team history**, not credentials; and the “context” of success is the coalition itself.

Political interpretation: ongoing joint work with Jorge Fábrega. Preliminary conclusions — the analysis continues.

¡Gracias! — Thank you



Project repository

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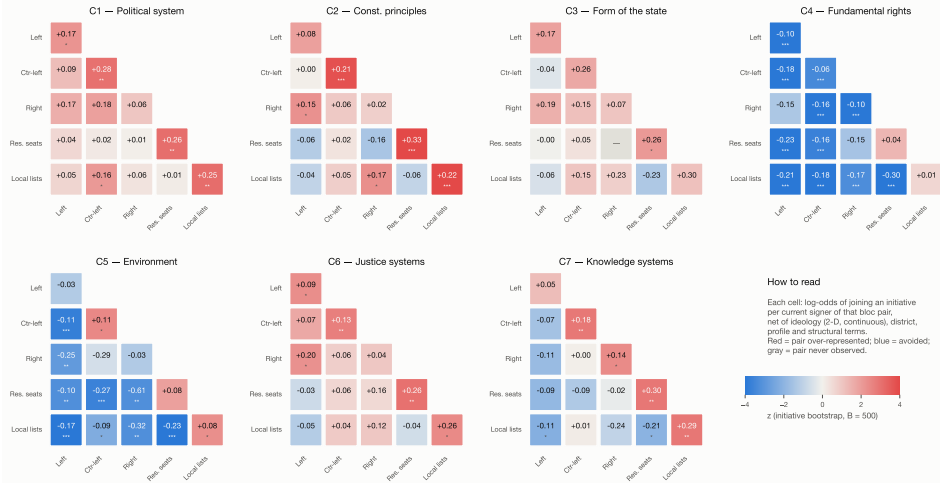
Backup

## Supporting material

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# Backup: the bloc mixing matrix (pure closure, no profile counters)

Who co-signs with whom, net of everything else — political-bloc mixing (bipartite ERGM, MPLE + bootstrap)



## Backup: the influence null, 3 windows $\times$ 7 exposure doses

| Window                 | Exposure definition                | $\hat{\lambda}$        | SE    | $p$            | N     |
|------------------------|------------------------------------|------------------------|-------|----------------|-------|
| Full                   | Accumulated since T0               | +0.007                 | 0.004 | 0.119          | 4,355 |
| Full                   | Last wave only                     | +0.007                 | 0.013 | 0.60           | 463   |
| Full                   | Last 2 waves                       | +0.003                 | 0.009 | 0.72           | 1,504 |
| Full                   | Last 3 waves                       | -0.002                 | 0.007 | 0.76           | 2,447 |
| Full                   | Decay $\lambda_w = 0.25/0.50/0.75$ | +0.008/ +0.008/ +0.007 | 0.004 | .051/.042/.078 | 4,355 |
| Full                   | Falsification: future exposure     | +0.010                 | 0.004 | 0.030          | 3,329 |
| Early ( $\leq$ Mar 31) | Accumulated                        | +0.005                 | 0.007 | 0.50           | 1,586 |
| Late ( $\geq$ Apr 1)   | Accumulated                        | +0.011                 | 0.005 | 0.018          | 2,769 |
| Late                   | Falsification: future exposure     | +0.010                 | 0.004 | 0.004          | 1,742 |

No  $\hat{\lambda}$  exceeds 0.011; the few significant late-window lags come with an equally significant falsification of the same size.

## Backup: how does each sector decide? (conditional logit, same data)

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|                        | Right   | Ctr-left | Left | Res. seats | Local lists |
|------------------------|---------|----------|------|------------|-------------|
| Distance in $\theta_1$ | -5.1*** | -2.8**   | -0.8 | +0.1       | -6.1***     |

- The complementary lens: the conditional logit conditions within each initiative (menu) and can split by the *decider's* bloc: the right almost never crosses ideological distance; the left crosses it all.
- The hybrid ERGM and the conditional logit agree on the substantive ranking — the tool changes what is visible, not what matters.